

> **# Reset maple instance**

> restart;
> with(linalg):

> **# Set up input vectors representing 2-D inviscid Euler equations**

> W:=array(1..4, [rho, u, v, p]);
> U:=array(1..4, [rho, rho*u, rho*v, (p/(g-1)) + (rho * (u^2 + v^2))/2]);
> F:=array(1..4, [rho*u, rho*u^2+p, rho*u*v, rho*u*((g*p)/((g-1)*rho) + (u^2+v^2)/2)]);
> G:=array(1..4, [rho*v, rho*u*v, rho*v^2 + p, rho*v*((g*p)/((g-1)*rho) + (u^2+v^2)/2)]);

$$W := \begin{bmatrix} \rho & u & v & p \end{bmatrix}$$

$$U := \begin{bmatrix} \rho & \rho u & \rho v & \frac{p}{g-1} + \frac{\rho(u^2 + v^2)}{2} \end{bmatrix}$$

$$F := \begin{bmatrix} \rho u & \rho u^2 + p & \rho u v & \rho u \left(\frac{g p}{(g-1) \rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \end{bmatrix}$$

$$G := \begin{bmatrix} \rho v & \rho u v & \rho v^2 + p & \rho v \left(\frac{g p}{(g-1) \rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \end{bmatrix} \quad (1)$$

> **# Compute Jacobians and other matrix quantities**

> dUdW:=array(1..4,1..4):
> dUdW:=jacobian(U,W);

$$dUdW := \begin{bmatrix} 1 & 0 & 0 & 0 \\ u & \rho & 0 & 0 \\ v & 0 & \rho & 0 \\ \frac{u^2}{2} + \frac{v^2}{2} & \rho u & \rho v & \frac{1}{g-1} \end{bmatrix} \quad (2)$$

> dWdU:=array(1..4,1..4):
> dWdU:=inverse(dUdW);

(3)

$$dWdU := \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{u}{\rho} & \frac{1}{\rho} & 0 & 0 \\ -\frac{v}{\rho} & 0 & \frac{1}{\rho} & 0 \\ \frac{(u^2 + v^2)(g-1)}{2} & -u(g-1) & -v(g-1) & g-1 \end{bmatrix} \quad (3)$$

```
> dFdW:=array(1..4,1..4):
dFdW:=jacobian(F,W);
```

$$dFdW := \begin{bmatrix} u, \rho, 0, 0, \\ u^2, 2\rho u, 0, 1, \\ uv, \rho v, \rho u, 0, \\ u \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) - \frac{ugp}{\rho(g-1)}, \rho \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) + \rho u^2, \\ \rho uv, \frac{ug}{g-1} \end{bmatrix} \quad (4)$$

```
> dGdW:=array(1..4,1..4):
dGdW:=jacobian(G,W);
```

$$dGdW := \begin{bmatrix} v, 0, \rho, 0, \\ uv, \rho v, \rho u, 0, \\ v^2, 0, 2\rho v, 1, \\ v \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) - \frac{vgp}{\rho(g-1)}, \rho uv, \rho \left(\frac{gp}{(g-1)\rho} + \frac{u^2}{2} + \frac{v^2}{2} \right) \\ + \rho v^2, \frac{vg}{g-1} \end{bmatrix} \quad (5)$$

```
> dFdU:=array(1..4,1..4):
> dFdU:=evalm(dFdW&*dWdU):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do dFdU[i,j]:=
factor(simplify(dFdU[i,j])) od: od:
> evalm(dFdU);
```

(6)

$$\left[\begin{bmatrix} 0, 1, 0, 0 \end{bmatrix}, \right. \quad (6)$$

$$\left[-\frac{3}{2} u^2 + \frac{1}{2} u^2 g + \frac{1}{2} g v^2 - \frac{1}{2} v^2, -u(g-3), -v(g-1), g-1 \right],$$

$$\left[-uv, v, u, 0 \right],$$

$$\left[\frac{u(g^2 \rho u^2 + g^2 \rho v^2 - 3 \rho u^2 g - 3 \rho v^2 g + 2 \rho u^2 + 2 \rho v^2 - 2 g p)}{2 \rho (g-1)}, \right.$$

$$\left. -\frac{2 g^2 \rho u^2 - 5 \rho u^2 g - \rho v^2 g + 3 \rho u^2 + \rho v^2 - 2 g p}{2 \rho (g-1)}, -uv(g-1), u g \right]$$

> dGdU:=array(1..4,1..4):

> dGdU:=evalm(dGdW&*dWdU):

> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do dGdU[i,j]:=factor(simplify(dGdU[i,j])) od: od:

> evalm(dGdU);

$$\left[\begin{bmatrix} 0, 0, 1, 0 \end{bmatrix}, \right. \quad (7)$$

$$\left[-uv, v, u, 0 \right],$$

$$\left[-\frac{3}{2} v^2 + \frac{1}{2} u^2 g + \frac{1}{2} g v^2 - \frac{1}{2} u^2, -u(g-1), -v(g-3), g-1 \right],$$

$$\left[\frac{v(g^2 \rho u^2 + g^2 \rho v^2 - 3 \rho u^2 g - 3 \rho v^2 g + 2 \rho u^2 + 2 \rho v^2 - 2 g p)}{2 \rho (g-1)}, -uv(g \right.$$

$$\left. -1), -\frac{2 g^2 \rho v^2 - \rho u^2 g - 5 \rho v^2 g + \rho u^2 + 3 \rho v^2 - 2 g p}{2 \rho (g-1)}, v g \right]$$

> # Compute matrices A and B

> A:=array(1..4,1..4):

> A:=simplify(evalm(dWdU&*dFdU&*dUdW));

$$A := \begin{bmatrix} u & \rho & 0 & 0 \\ 0 & u & 0 & \frac{1}{\rho} \\ 0 & 0 & u & 0 \\ 0 & g p & 0 & u \end{bmatrix}$$

(8)

> B:=array(1..4,1..4):

> B:=simplify(evalm(dWdU&*dGdU&*dUdW));

$$B := \begin{bmatrix} v & 0 & \rho & 0 \\ 0 & v & 0 & 0 \\ 0 & 0 & v & \frac{1}{\rho} \\ 0 & 0 & g\rho & v \end{bmatrix} \quad (9)$$

> # Outline algebraic assumptions

```
> assume(g, real);
> assume(g > 0);
> assume(rho, real);
> assume(rho > 0);
> assume(a, real);
> assume(a > 0);
> assume(p, real);
> p:=rho*a^2/g;
```

$$p := \frac{\rho a^2}{g} \quad (10)$$

> # Compute eigenvalues of A and B

```
>
# A and script A are similar matrices, so we can show that they have the same
eigenvalues
```

```
I4:=array(1..4,1..4, identity):
> char_poly1A:=factor(simplify(det(A-lambda*I4))):
> eigenvals1A:=solve(char_poly1A=0, lambda);
> char_poly2A:=factor(simplify(det(dFdU-lambda*I4))):
> eigenvals2A:=solve(char_poly2A=0, lambda);
    eigenvals1A := u - a~, u + a~, u, u
    eigenvals2A := u - a~, u + a~, u, u \quad (11)
```

```
>
# The same is true of B and script B
```

```
> char_poly1B:=factor(simplify(det(B-lambda*I4))):
> eigenvals1B:=solve(char_poly1B=0, lambda);
> char_poly2B:=factor(simplify(det(dGdU-lambda*I4))):
> eigenvals2B:=solve(char_poly2B=0, lambda);
    eigenvals1B := v - a~, v + a~, v, v
    eigenvals2B := v - a~, v + a~, v, v \quad (12)
```

> # Find eigenvectors using the eigenvalues

```
> # initialize arrays to hold the eigenvectors
rightAeigenvectors:=array(1..4,1..4):
rightBeigenvectors:=array(1..4,1..4):
```

```
# begin with the eigenvectors of A
```

```
>
# eigenvalue 1 of A (eigenvector 1)
AminLam1:=array(1..4,1..4):
> AminLam1:=evalm(A-evalm((eigenvals1A[1])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> r1A:= nullspace(AminLam1)[1];
```

$$r1A := \begin{bmatrix} -\frac{\rho}{a} & 1 & 0 & -a\rho \end{bmatrix} \quad (13)$$

```
>
# eigenvalue 2 of A (eigenvector 2)
AminLam1:=evalm(A-evalm((eigenvals1A[2])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> r2A:=nullspace(AminLam1)[1];
```

$$r2A := \begin{bmatrix} 1 & \frac{a}{\rho} & 0 & a^2 \end{bmatrix} \quad (14)$$

```
>
# eigenvalue 3 of A (eigenvectors 3 and 4 since this is a repeated eigenvalue)
AminLam1:=evalm(A-evalm((eigenvals1A[3])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> nullspaceA3:=nullspace(AminLam1):
> r3A:=nullspaceA3[1];
```

$$r3A := \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad (15)$$

```
> r4A:=nullspaceA3[2];
```

$$r4A := \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \quad (16)$$

```
>
# eigenvalue 4 of A (should have an identical output of eigenvalue 3)
AminLam1:=evalm(A-evalm((eigenvals1A[4])*I4)):
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do AminLam1[i,j]
:=simplify(subs(p=rho*a^2/g, AminLam1[i,j])) od: od:
> nullspace(AminLam1);
```

$$\left\{ \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \right\} \quad (17)$$

```
>
# now compute the eigenvectors of B
```

```
# eigenvalue 1 of B (eigenvector 1)
```

```
BminLamI:=array(1..4,1..4):
```

```
> BminLamI:=evalm(B-evalm((eigenvals1B[1])*I4)):
```

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
```

```
> r1B:=nullspace(BminLamI)[1];
```

$$r1B := \begin{bmatrix} -\frac{\rho \sim}{a \sim} & 0 & 1 & -a \sim \rho \sim \end{bmatrix} \quad (18)$$

```
>
```

```
# eigenvalue 2 of B (eigenvector 2)
```

```
BminLamI:=evalm(B-evalm((eigenvals1B[2])*I4)):
```

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
```

```
> r2B:=nullspace(BminLamI)[1];
```

$$r2B := \begin{bmatrix} 1 & 0 & \frac{a \sim}{\rho \sim} & a \sim^2 \end{bmatrix} \quad (19)$$

```
>
```

```
# eigenvalue 3 of B (eigenvectors 3 and 4 since this is a repeated eigenvalue)
```

```
BminLamI:=evalm(B-evalm((eigenvals1B[3])*I4)):
```

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
```

```
> r3B:=nullspace(BminLamI)[1];
```

$$r3B := \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad (20)$$

```
> r4B:=nullspace(BminLamI)[2];
```

$$r4B := \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \quad (21)$$

```
>
```

```
# eigenvalue 4 of B (should have an identical output of eigenvalue 3)
```

```
BminLamI:=evalm(B-evalm((eigenvals1B[4])*I4)):
```

```
> for i from 1 to 4 by 1 do for j from 1 to 4 by 1 do BminLamI[i,j]
:=simplify(subs(p=rho*a^2/g, BminLamI[i,j])) od: od:
```

```
> nullspace(BminLamI);
```

$$\left\{ \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \right\} \quad (22)$$

```
>
```

Check the eigenvalue problem residuals

```
> # check residuals for eigenvalue eigenvector pairs of matrix A
```

```
res:=array(1..4):
```

```
> res0:=evalm((A&*r1A)-eigenvals1A[1]*r1A):
```

```
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])
)) od:
```

```
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (23)$$

```
> res0:=evalm((A&*r2A)-eigenvals1A[2]*r2A):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (24)$$

```
> res0:=evalm((A&*r3A)-eigenvals1A[3]*r3A):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (25)$$

```
> res0:=evalm((A&*r4A)-eigenvals1A[4]*r4A):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (26)$$

```
>
# check residuals for eigenvalue eigenvector pairs in matrix B
```

```
> res0:=evalm((B&*r1B)-eigenvals1B[1]*r1B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (27)$$

```
> res0:=evalm((B&*r2B)-eigenvals1B[2]*r2B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (28)$$

```
> res0:=evalm((B&*r3B)-eigenvals1B[3]*r3B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (29)$$

```
> res0:=evalm((B&*r4B)-eigenvals1B[4]*r4B):
> for i from 1 to 4 by 1 do res[i]:=factor(simplify(expand(res0[i])) od:
> evalm(res);
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \quad (30)$$

Compute the eigenvectors for the left eigenproblem

```
> # make right eigenvector matrix for A
```

```
RA:=array(1..4,1..4, sparse):
```

```
> for i from 1 to 4 by 1 do RA[i,1]:=simplify(r1A[i]) od:
```

```
> for i from 1 to 4 by 1 do RA[i,2]:=simplify(r2A[i]) od:
```

```
> for i from 1 to 4 by 1 do RA[i,3]:=simplify(r3A[i]) od:
```

```
> for i from 1 to 4 by 1 do RA[i,4]:=simplify(r4A[i]) od:
```

```
>
```

```
# make the left eigenvector matrix for A by inverting the right eigenvector matrix
```

```
> LA:=array(1..4,1..4, sparse):
```

```
> LA:=inverse(RA):
```

```
> l1A,l2A,l3A,l4A:= vector(4),vector(4),vector(4),vector(4):
```

```
> for i from 1 to 4 by 1 do l1A[i]:=simplify(LA[1,i]) od:
```

```
> for i from 1 to 4 by 1 do l2A[i]:=simplify(LA[2,i]) od:
```

```
> for i from 1 to 4 by 1 do l3A[i]:=simplify(LA[3,i]) od:
```

```
> for i from 1 to 4 by 1 do l4A[i]:=simplify(LA[4,i]) od:
```

```
>
```

```
> evalm(l1A);
```

```
> evalm(l2A);
```

```
> evalm(l3A);
```

```
> evalm(l4A);
```

$$\begin{bmatrix} 0 & \frac{1}{2} & 0 & -\frac{1}{2a\tilde{\rho}} \end{bmatrix}$$

$$\begin{bmatrix} 0 & \frac{\rho}{2a} & 0 & \frac{1}{2a^2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{a^2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$$

(31)

```
> # make right eigenvector matrix for B
```

```
RB:=array(1..4,1..4, sparse):
```

```
> for i from 1 to 4 by 1 do RB[i,1]:=simplify(r1B[i]) od:
```

```
> for i from 1 to 4 by 1 do RB[i,2]:=simplify(r2B[i]) od:
```

```
> for i from 1 to 4 by 1 do RB[i,3]:=simplify(r3B[i]) od:
```

```
> for i from 1 to 4 by 1 do RB[i,4]:=simplify(r4B[i]) od:
```

```
>
```

```
> # make the left eigenvector matrix for B by inverting the right eigenvector
```

